

W1
GAA

* Financial statement of a company include.

- ↳ Balance sheet
- ↳ Profit/loss statements
- ↳ cash flow statements

* Production is

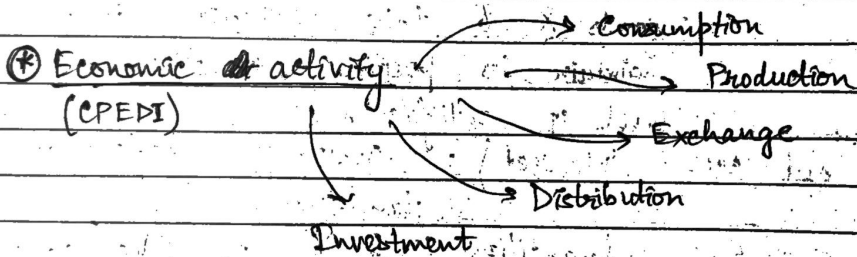
- ↳ a process of value addition.

* Utility is a measure of the satisfaction received by a customer, when a good or service is consumed.

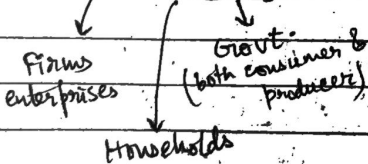
(measured in terms of utils)

* Consumption is economic activity concerned with use of goods

* Demand is created by consumption.



④ 3 agents of economy

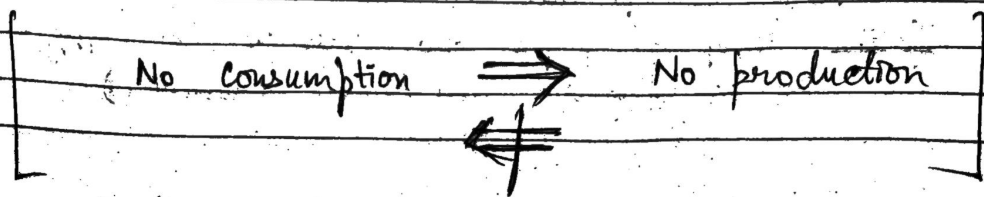


* How are the 5 activities interrelated?

(Ans) When consumers consume, producers produce, and resources are exchanged among them and this distribution of the exchange determines the investment and in turn determines the production capacity.

<u>Micro Economies</u>	<u>Macro Economies</u>
Economic activities at the level of individuals, households, firms.	Economic activities at the level of state, nation, global, industrial, sectorial in aggregated terms.

- * Notion of utility: Utility is implicit to a commodity. Any commodity without utility value will have no takers.
- * Sampling is a technique of selecting individual members or a subset of the population to make statistical inferences from them.
- * Stratified sampling is a method where your sampling is modeled based on different strata such as income, region, occupation, etc.
- * Sampling data & sample size should be representative of the entire population.
- consumption basket: If we categorize all consumption into groups for a certain period of time (say a day, a week, a month or so), we term them as consumption baskets.
- Consumption theory: A consumer has limited means/income while their wants could be unlimited. It is a theory of formulation or generalization of standard relationships, that explain how consumers tend to behave.



- Time-series data: Data points typically consist of successive measurements made from the same source over a time interval and are used to track change over time.
- Cross-sectional data: Provides a snapshot of data at a certain instance or point of time.
- Panel data: It is a type of data which is pooled from both time series & cross-sectional data.

W2

Consumer Behaviour Approaches

- cardinal (MUs)
- Ordinal (ICs)

cardinal (Quantitative)

TU: Total utility derived from consumption of a commodity

MU: Marginal utility

$$MU = \frac{\Delta TU}{\Delta Q}$$

$\nearrow$ change in utility
 $\searrow$ change in quantity

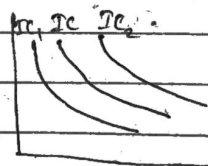
At TU_{max} , $MU > 0$
 $TU \downarrow$, $MU < 0$ (ve)

$TU \uparrow$, $MU > 0$ (ve)

Ordinal (Qualitative)

IC is a representation that gives ~~same~~ a combo of commodities that provide the same level of satisfaction for the customer.

Higher IC $\Rightarrow$ higher level of satisfaction
 Lower IC $\Rightarrow$ lower level of satisfaction.



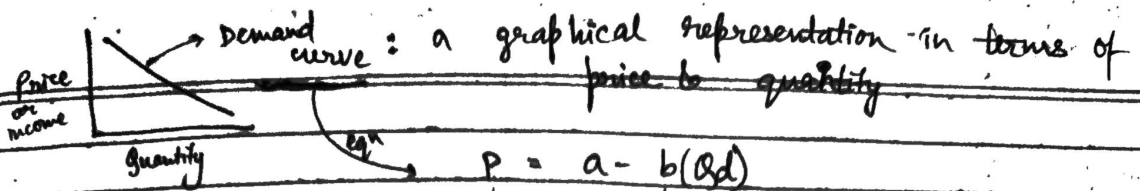
$$Sat(IC_1) < Sat(IC) < Sat(IC_2)$$

* A market is a group of buyers & sellers of a particular good or service.

Buyers: Group of people who determine the demand of a product

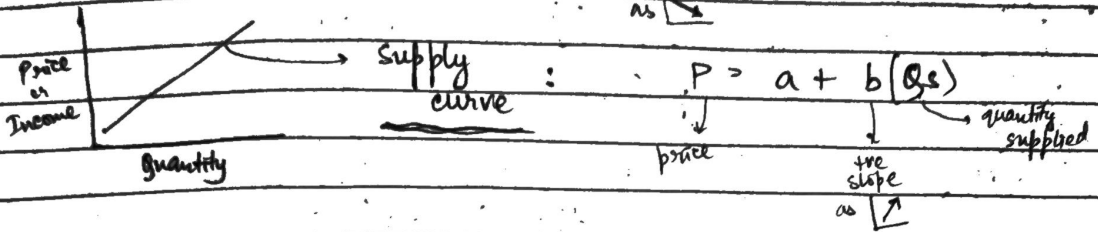
Sellers: Group of people who determine the supply of a product (good/service/commodity).

- Law of demand: Price (commodity) $\uparrow$; demand $\downarrow$ and vice-versa



$$P = a - b(Q_d)$$

$\downarrow$ price $\downarrow$ quantity demanded
 -ve slope as $\downarrow$

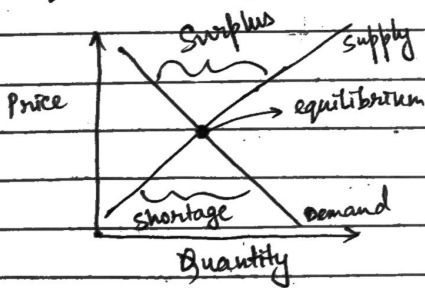


$$P = a + b(Q_s)$$

$\downarrow$ price $\downarrow$ quantity supplied
 +ve slope as $\uparrow$

- * **Shifts in demand curve** } Reasons: Income / Price / Season / Price of complements / Price of Tastes / substitutes / Advertisements / Fashion so on & on -
 - Right shift $\Rightarrow \uparrow$ demand
 - Left shift $\Rightarrow \downarrow$ demand

* **Market equilibrium**: When quantity demanded & quantity supplied becomes equal, it is said to be at market equilibrium (situation when demand-supply is balanced)



Surplus: Excessive supply
 Shortage: Excessive demand

* **Expectations: Income**

* **Normal goods**: Commodities whose demand $\uparrow$ when income $\uparrow$

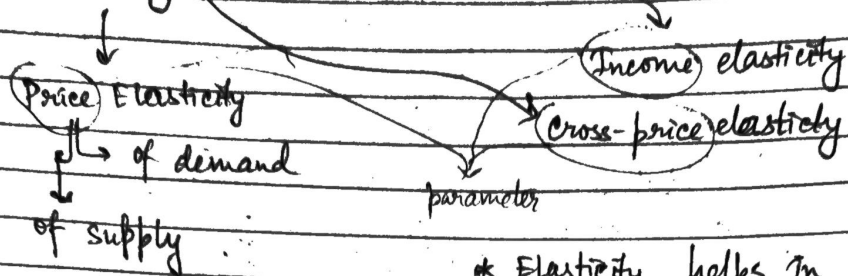
* **Inferior goods**: Commodities whose demand $\downarrow$ when income $\uparrow$

* **Substitutes**: ~~Two~~ A pair of commodities are substitutes if for an increase in demand of one leads to a decrease " " the other.
 (e.g. train ticket, plane ticket)

* **Complements**: A pair of commodities are complements if an increase in demand of one causes " " of the other as well

Law of supply and Demand: The price of any good adjusts to balance the quantity supplied and quantity demanded for that particular good.

Elasticity



* Elasticity helps in understanding the variation in demand.

$$\text{Elasticity} = \left(\frac{\Delta Q}{Q_{\text{initial}}} \right) / \left(\frac{\Delta \text{Para}}{\text{Para}_{\text{initial}}} \right)$$

→ quantity
→ parameters

* PED (Price Elasticity of Demand)

$$\text{PED} = \left| \frac{(\Delta Q / Q_{\text{initial}})}{(\Delta P / P_{\text{initial}})} \right|$$

→ PED > 1 : Elastic

→ PED < 1 : Inelastic

→ Unit PED = 1 : Unit Elasticity

→ PED = 0

Perfectly inelastic

→ PED = ∞

Perfectly elastic

* IED (Income elasticity of demand)

$$\text{IED} = \left(\frac{\Delta Q}{Q_{\text{initial}}} \right) / \left(\frac{\Delta \text{Inc}}{\text{Inc}_{\text{initial}}} \right)$$

⇒ 0 < IED [ve] : Normal goods

↳ 0 < IED < 1 : Necessity

↳ IED > 1 : Luxury

⇒ IED < 0 [-ve] : Inferior goods.

* Elastic : When demand (or supply) responds substantially to ~~small~~ (very much) change in price.

* Inelastic : When demand (or supply) responds meagerly to change in price (slightly).

* PEOS (Price elasticity of supply)

→ $PEOS > 1$: Elastic

→ $PEOS < 1$: Inelastic

→ $PEOS = 1$: Unit elasticity

* Cross-price elasticity : (CPEOD) : $\frac{\left(\frac{\Delta Q_1}{Q_{1, \text{initial}}}\right)}{\left(\frac{\Delta Q_2}{Q_{2, \text{initial}}}\right)}$

Q_1 : First commodity
 Q_2 : 2nd " quantity

⊛ 2 broad agents of Market : ① Consumer
② Producer

- Historical cost : Actual cost of an asset
- Replacement cost : Present cost of the asset.
- Depreciation reserve : A reserve that accommodates certain specific amount into this reserve per year to avoid any additional burden on the firm in future (to meet the replacement cost)
- Fixed cost : Any cost that are unaffected by the company's output ^{or} sales.
- Variable cost : Any cost that gets affected by the company's output ^{or} sales.
- Real cost : Cost of any materialistic things required
- Prime cost : Cost incurred in terms of material and labour involved ~~excluding~~ excluding fixed cost.

* Production function gives relationship between inputs used in the production and the max output production with given period of time and with given period of technology.

(KLEM : Capital, Labour, Energy, Material)

$$Q = f(x_1, x_2, \dots, x_n) \quad \text{where } x_i \text{ : inputs used in production} \\ i = 1(1)n$$

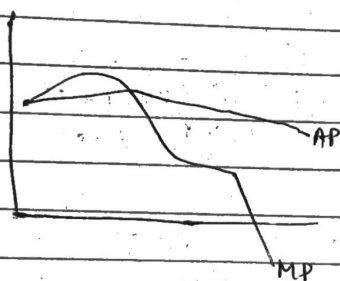
Short-run analysis

$$Q = f(X, Y) \quad \text{where } X : \text{labour, } Y : \text{capital}$$

$$\text{Marginal product (MP)} = \frac{\Delta Q}{\Delta X}$$

$$\text{Average product (AP}_x\text{)} = Q/X$$

Total product (TP)



$$MP > AP \Rightarrow AP \uparrow$$

$$MP = AP \Rightarrow AP_{\max}$$

$$MP < AP \Rightarrow AP \downarrow$$

* Cobb-Douglas production function

It is a 2-input exp function of capital & labour

$$Q = aL^b K^c$$

$$(b+c) > 1 \quad \text{IRTS}$$

$$(b+c) = 1 \quad \text{CRTS}$$

$$(b+c) < 1 \quad \text{DRTS}$$

* Total Revenue : Total Quan * Price per Quantity

* Average " : Total rev / Total Quan produced

* Marginal " : $\Delta \text{Revenue} / \Delta \text{Total Quan}$

Market Skimming

low volume
high price
(Eg: Playstation)

Value Pricing

Based on survey
products are
categorized &
accordingly priced.

Loss Leader Pricing

~~loss~~ sell at
reduced cost to
encourage large
volume sales

Psychological Pricing

Rs 9.99
Rs 10.99

Growing rate pricing

Constantly monitor
rival's pricing
→ Match the price
or
cut the price

Tender rate pricing

low-cost, high value
delivery model
(Typical govt tenders)

Price Discrimination

Same product
sold at different
prices to different
people
(Airline tickets)

Peak-load pricing

(A variant of price
discrimination)
Eg: metro prices at
peak hours v.s.
non-peak hours

Cross-price subsidization

If a set of customers
are unable to pay for
service 1, then price
of service 1 ↓ &
price of some
service 2 for another
set of customers is
increased (Eg:
Train tickets; SL, AC, ...)

Penetration Pricing

Pricing set
to penetrate the
market
(Jio Fiber) Initially
very very low cost
focusing on high vol
sales (long term strategy)

Cost-plus pricing

Pricing model
used to maximize
the rates of return
for the companies
 $CP + \text{Markup (MU)}$
= Selling price (SP)

Contribution pricing

Contribution
 $= MU + FC$

$$CP = FC + VC$$

fixed variable

Marginal cost pricing

$MC = \frac{\Delta \text{Total Cost}}{\Delta \text{Output}}$
covers cost of
producing 1 extra
unit or 1 fewer units

Absorption pricing

covers variable
costs & fixed costs

Target pricing

Target price X
to reach X, vary MUs

$$MU = \frac{\text{Profit}}{\text{Cost}} \times 100$$

+ MU

Destructive pricing

prevents new
entrants
Illegal Δ

- The shape of average total cost curve will be U-shaped (ATC)
- In the economics scale region, $ATC \downarrow$
- $\exists$ 1-1 relationship between cost in the constant returns to scale region
- The plot between ATC & Quantity of prod per time period generates the average cost curves.

W3 Analysis of firm performance - Key ratios

Reason of financial analysis

- * Financial analysis of the firm helps in finding its own strengths and weaknesses
- * helps in monitoring the key indicators of the firm

Ratio Analysis $\rightarrow$ helps firms to ~~analyse~~ evaluate performance, do structural analysis, provide useful insights

$\downarrow$
to study efficiency of operations, its risks

- TCA : Total current assets
- Cash
- CE : Cash Equivalents
- MS : Marketable securities
- NAR : Net accounts receivable
- PE : Prepaid expenses
- SI : Stock inventory
- CL : current liabilities

$$* \quad TCA = (\text{Cash} + CE) + MS + NAR + PE + SI$$

$$* \quad \text{Current Assets} = (\text{Cash} + CE) + MS + NAR + SI$$

$$> TCA - PE$$

* Quick Assets

= Current Assets - SI

= TCA - SI - PE

Quick ratio: Measures ability of a company to meet its short-term obligations with its most liquid assets.

(Acid test ratio)

(meet any urgent requirement)

$$QR = \frac{\text{Quick Assets}}{CL}$$

1:1 $\equiv$ Ideal ratio $QR > 1$

3:1 $\equiv$ Healthy ratio $QR \geq 3$

0.5:1 $\equiv$ Unhealthy ratio $QR \leq 0.5$

Current ratio: Liquidity ratio that measures the ability of a firm to meet its current obligations

$$CR = \frac{\text{Current Assets}}{CL}$$

$CR \geq QR$

Too high / Too low $\equiv$ Problematic

(1.5:1) $\equiv$ Ideal ratio

* Earnings per share ratio: Gives a signal or indication for potential investors.

$$\text{Ratio} = \frac{\text{Profit after tax}}{\text{no of shares}}$$

* Price earnings ratio

$$= \frac{\text{market price}}{\text{earnings/share per}}$$

* Dividend ratio

$$= 100 \left(\frac{\text{ordinary share dividend}}{\text{market price}} \right)$$

↑ the better

* Gearing ratio

$$= \frac{\text{long term loans}}{\text{capital employed}} \times 100$$

- Gross Profit = $\frac{\text{Total revenue}}{\text{Total revenue}} - \text{VC}$

- Net profit = $\frac{\text{Total revenue}}{\text{Total revenue}} - \text{VC} - \text{FC}$

- Gross profit margin = $\frac{\text{gross profit}}{\text{total revenue}} \times 100$

- Net profit margin = $\frac{\text{net profit}}{\text{total revenue}} \times 100$

* ROCE Return of capital Employed

$$= \frac{\text{Profit}}{\text{capital employed}} \times 100$$

- Asset turnover ratio : $\frac{\text{Sales turnover}}{\text{assets employed}}$

[Capital Employed : TCA]

- Stock turnover ratio : $\frac{\text{costs of goods sold}}{\text{stocks expressed as times per year}}$

- Debtor days = $\frac{\text{Debtor}}{\text{Sales turnover}} \times 365$

Herfindahl Index

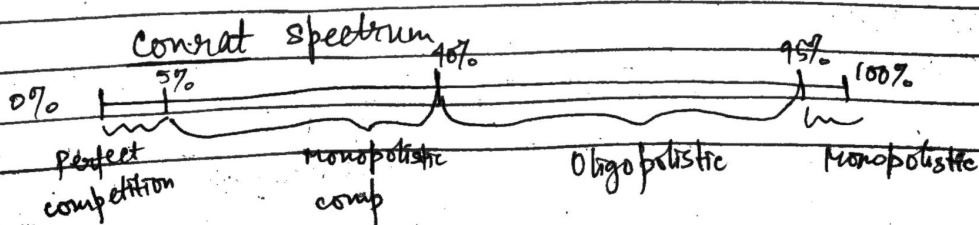
► Concentration ratio is the total percentage ^{output} of top 4 largest firms in the industry.

↳ Compute % output of each firm

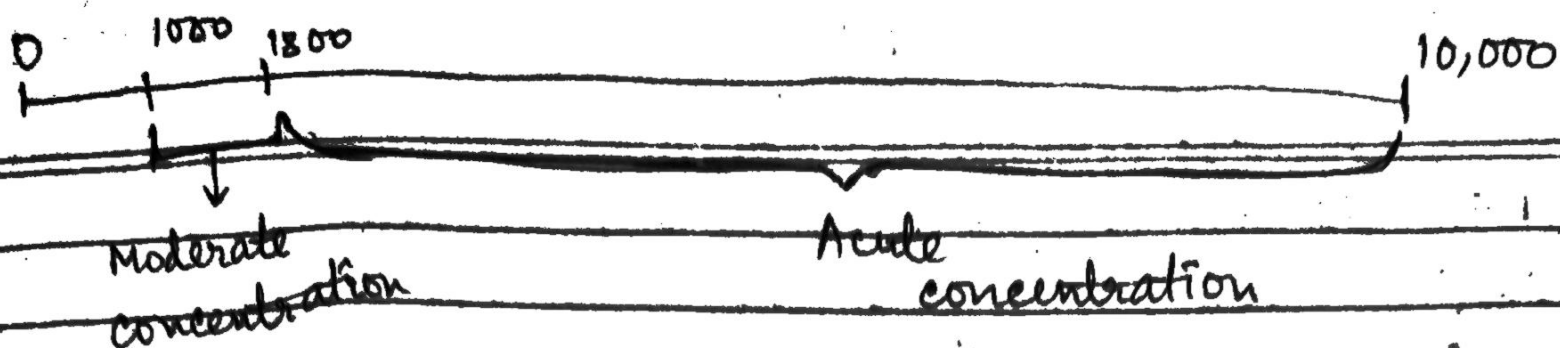
↳ Square them

↳ Add them to get the Herfindahl index

↳ Interpret it on H.I spectrum



HI index spectrum



- MNC : any company with $>10\%$ equity and with operations in more than 1 country

* If production ~~capa~~ volume is increased then

- ↳ capacity utilization increases
- ↳ ROCE is increased.

* Principal source for most of the basic statistics of the industrial sector is annual survey of industries

Private sector industry	Public Sector Industry	Joint sector Industry	Cooperative sector industry
Tata Steel	BHEL	Maruti Udyog	Amul

- ① While conducting National Sample Survey, the following can affect the data.

- Recall bias

- Timing of survey

$$PEOD = \left| \frac{\Delta Q/Q_c}{\Delta P/P_c} \right|$$

- ② Utility drives consumer's choices

$PEOD < 1$: Inelastic ≤ 0 perfectly in

> 1 : elastic ∞ perfectly (out)

$= 1$: Unit

Necessity is an inelastic

- ③ Industry is a collection of firms

- ④ Volume is prioritized over price ~~where~~ to decide which is to be kept closer to packaging area?

$$ROCE = \frac{\text{Profit}}{\text{Capital employed}} * 100$$

→ calculating channel effectiveness? not relevant
campus recruitment } to internal sourcing process

- A/B testing after deploying the model in realworld

M3, 4, 7
F2

OEE (Overall Equipment Effectiveness Calculation)

$$OEE = \left(\frac{\text{Availability Rate}}{\text{Rate}} \right) * \left(\frac{\text{Performance Rate}}{\text{Rate}} \right) * \left(\frac{\text{Quality Rate}}{\text{Rate}} \right)$$

$\rightarrow = \frac{\text{Actual speed}}{\text{Design speed}}$

Para	Wkt.	A.R = $\frac{b}{a}$
AR (PPH)	80 hr	$a = 80$
AR (LOT)	8 hr	$b = 72$
PR (Designed Gear)	55/Hr	
PR (Actual Gear)	50/Hr	$P.R = \frac{d}{c}$
QR (Total prods)	3600 units	$d = 50$
QR (Scrap)	137 units	$c = 55$

$$Q.R = \frac{f}{e}$$

$$f = 3600 - 137$$

$$= 3463$$

$$e = 3600$$

$$= 50/55 = 0.9$$

$$\therefore QR = \frac{3463}{3600} = 0.96$$

$$\therefore OEE = (0.9)(0.9)(0.96) = 0.785 \approx 0.79$$

* Open stock is never negative

* Increase in (Scrap; Direct Material Cost; Direct Labour Cost; Shipping Cost) reduces Gross Margin.

